

Efficiency measurement of Indian high courts using DEA: A policy perspective

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Abstract

The Indian judicial system is extensive and plagued by several issues, especially high pendency levels across all levels. The present work has two goals. Firstly, it aims to measure the efficiency of Indian high courts using Data Envelopment Analysis (DEA). Secondly, it studies the impact of including pending cases on judicial efficiency. The first DEA model takes only judicial resources, namely, number of judges and staff members as inputs, and two outputs, viz., number of civil and criminal cases disposed. Since further data analysis reveals a role for using caseload as an input, two more DEA models are developed to incorporate these factors. The first includes the number of civil and criminal cases instituted during the year as inputs, and the second incorporates the effect of both incoming and pending cases. Models 2 and 3 help to distinguish between the courts that are efficient with respect to incoming cases, and those that are able to efficiently manage their total workload. Results identify the specific courts that are efficient in disposing cases, including the effect of their high volume. They point to policy imperatives and overall peer learning, as well as for specific aspects such as dealing with high pendency or fresh institution of cases. Finally, a comparison between these models can help the judicial officials of inefficient courts develop reforms with specific aims such as reducing backlog of cases, matching outflow and inflow of cases, or in some cases, both.

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1. Introduction

India is an emerging economy with a population of over 1.3 billion. In terms of the World Bank's quality of judicial processes index, India scores 10.5 on a scale from 0 to 18. It costs nearly 31% of the claim value and 1445 days on an average to resolve a commercial dispute in Indian courts of first instance.

Justice is guaranteed by the Indian State to all citizens under the Preamble and Articles 14, 19, 21, 32, 226 of the Constitution of India. The timely dispensation of justice is a constitutional obligation of the State in light of the Directive Principles of State Policy articulated in Articles 38(1), 39 and 39A of the Constitution of India. However, Indian courts are known for their inability to meet the public's demand, slow resolution of cases, and the building backlog (Devulapalli & Padmanabhan, 2019; Ministry of Finance, 2019). The conventional wisdom believes that the primary reason is a severe shortage of judges and inability to fill judicial vacancies (Deshpande, 2019; Mathur & Mandal, 2018; Thankur, 2019).

Additionally, however, other aspects need to be considered. For example, Buscaglia and Dakolias (1999) finds that nations which invest more in technology and infrastructure show improvements in clearance rates and duration of cases as compared to those that focus on increasing judicial strength and salaries alone. Some studies such as Walsh (2008) argue that India needs to move from talking about judicial numbers to determining judicial productivity. Table 1 presents the Indian judicial system in numbers as on 31 July, 2019. There are around 18 judges per million population in India, against a recommended figure of 50 per million in Law Commission of India (1987b). The judicial vacancies across high courts for the year 2017–18 are presented in Fig. 1. The sanctioned judge strength of a court is the sum of the working judge strength and vacancies. For example, Delhi has 25 judge vacancies and a working strength of 35, against a sanctioned judicial strength of 60. These numbers are important and concerns therein need to be addressed.

There exists a gap in terms of analysing the performance of different levels of Indian judiciary and suggesting reforms based on the results of performance evaluation. The present work aims to study the relative efficiency of Indian high courts using Data Envelopment Analysis (DEA). The efficiency of high courts of India has not been studied and published so far to the best of our knowledge. Thus, this study is the first of its kind that analyzes their efficiency (based on DEA) and ranks the courts accordingly. The study also aims to analyse the impact of workload on the judicial efficiency of the courts. We also try to determine if the court is efficient with respect to incoming cases or the overall caseload (sum of cases pending at the beginning of the year and

Table 1
Indian judicial system in numbers (as on 31 July, 2019).

	High courts	District courts
Number of pending cases		
• Civil	1854267	8807018
• Criminal	1196670	21888363
• Writ	1261949	–
• Total	4312886	30695381
Pending cases for over a decade	821912	2456783
Sanctioned strength of judges	1079	22570
Average pending cases per judge	3997	1360
Sanctioned judges per million population		18

Source: National Judicial Data Grid, 31 July, 2019.

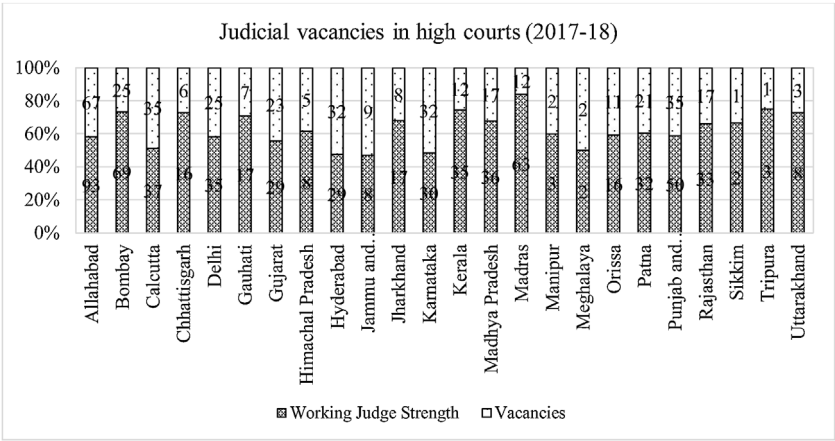


Fig. 1. Judicial vacancies in high courts (2017–18).

number of cases instituted during the year), and study the impact of including pending cases on efficiency. This would enable the decision makers in the judicial system understand the specific area where the court is lagging and needs attention. Accordingly, appropriate reforms can be developed and implemented.

The rest of this paper is organized in the following sections: Section 2 gives an overview of the Indian judicial system and possible impacts of a weak judiciary. Section 3 gives a review of literature on DEA and its applications. Section 4 presents the proposed DEA models and results. Section 5 presents the learnings and policy implications from this study. Section 6 concludes the paper with a summary and further extensions of this work.

2. Indian judicial system

The Indian judicial system has a pyramid-like structure: Supreme Court at the apex, followed by high courts and district courts. Currently, there are 25 high courts in India. Each high court has jurisdiction over a state, or a group of states and union territories. It has the power of superintendence over the courts and tribunals across the territories over which it exercises jurisdiction and makes decisions regarding the state judiciary in consultation with the state government. The lower judiciary consists of district and subordinate courts to deal with civil cases, and sessions and magisterial courts to deal with criminal cases. There are special courts to handle specific matters such as consumer courts, family courts, and fast track courts.

Indian courts have a common law system which allows flexibility in the interpretation of laws by different courts. Variations across state courts can arise due to differences in interpretations of rules and procedures, and administrative control of different state governments (Ahsan, 2013).

Clearance rate is defined as the ratio between number of cases disposed and number of cases instituted. The courts have performed well in 2017–18 in terms of clearance rate, with a national average of 0.88 (Fig. 2). However, a clearance rate above 1 is needed to enable the courts to clear some backlog each year. Only two courts have a clearance rate above 1 (Gauhati and Orissa). As per Ministry of Finance (2019), Indian high courts need 93 additional judges to achieve a clearance rate of 1, and 361 additional judges to clear the backlog in the next five years. However, these numbers don't incorporate the effects of the relative efficiencies of the courts. This study

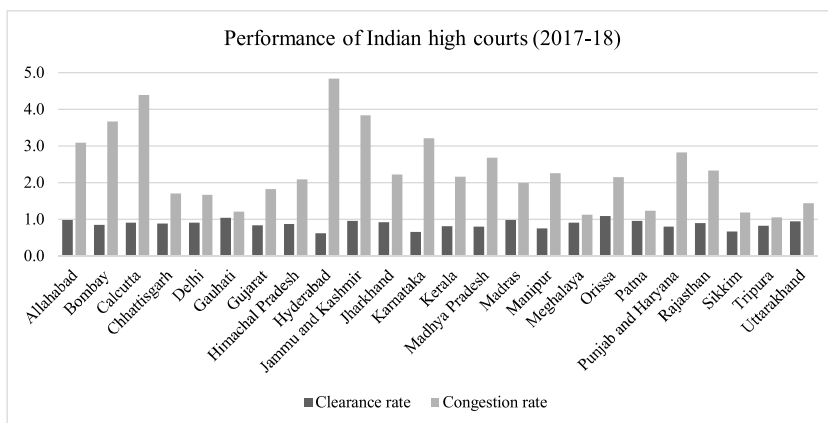


Fig. 2. Performance of high courts (2017–18).

aims to address this gap. Further, this study also indicates the role of congestion in the overall performance. Congestion rate is defined as the ratio between number of cases older than a year (pending cases at the beginning of the year) to the number of cases disposed (Micevska & Hazra, 2004). It indicates the number of years the court would take to clear its backlog based on current clearance rate. For example, Hyderabad and Calcutta are highly congested, with congestion rates of 4.8 and 4.4, respectively (Fig. 2).

Majority work on Indian high courts pertains to studying the impact of a weak judiciary, and not on evaluating their efficiency. Traditional measures such as percentage of pending trials (Amaral & Bandyopadhyay, 2015), caseload per judge (Micevska & Hazra, 2004), and average duration of cases (Chemin, 2009) are used as indicators of judicial performance. A country will experience high economic growth if its judicial system is able to protect the rights of its investors (Beck, Demirguc-Kunt, Levine, & Maksimovic, 2000; Haidar, 2009). The quality of the Indian judicial system has a significant impact on the economic growth, poverty rates, and agricultural output across the country (Chemin, 2009; Köhling, 2000). A propensity to engage in criminal activities also increases as a result of rising backlogs in high courts (Amaral & Bandyopadhyay, 2015).

India has a low settlement rate (over 90% cases require a court verdict, with only 5% cases being settled), and high adjournment rate not allowing for continuous trials (20–40 adjournments during the life of a case) (Walsh, 2008). Various reports by the Law Commission of India, Ministry of Law and Justice, India have focused on increasing strength of judges or filling vacant judgeships (Law Commission of India, 1979, 1987a, 1988), and emphasized that delays are on account of mismatch of demand and supply (Law Commission of India, 1978). However, simply increasing the number of judges to combat the backlog without considering other issues would result in increasing financial and administrative burden (George, 2014). Effective and relevant reforms need to be instituted to ensure improved judicial performance.

3. Review of literature

Various techniques such as Data Envelopment Analysis (DEA) and mathematical programming have often been used in public sector domains such as education (Bhatnagar & Bolia, 2019), evacuation planning (Lakshay & Bolia, 2019), judicial systems (Santos & Amado, 2014), and healthcare services (Nedelea & Fannin, 2013). DEA is a widely sought after tool to measure

the performance of organizational units. It is a non-parametric, linear programming technique, used to measure the relative efficiency of homogenous units, termed as decision making units (DMUs). DEA allows for multiple inputs and outputs with varying units of measurement. It does not require any a priori restrictions on the weights of the variables. Apart from measuring the relative efficiency of each DMU, DEA provides additional information for inefficient DMUs: efficient peers that act as role models, their relative weights, and slack and target values of the variables considered. Unlike parametric techniques, it does not require any assumptions about the production function, and the distribution of random error and inefficiency terms (Nissi & Rapposelli, 2010).

DEA was introduced in Charnes, Cooper, and Rhodes (1978). A mathematical model (known as CCR) is depicted to assess the relative efficiency of units. It may be input or output-oriented. The primal and dual forms of the model are also known as multiplier and envelopment forms, respectively. The multiplier form gives the pure weights of the variables. Pure weights represent the marginal rate of substitution and depend on the unit of measurement of the variables. Virtual weights are calculated by multiplying the value of the variables with the respective pure weight. Virtual weights can be compared to get the relative contribution of each variable to the unit's efficiency (Santos & Amado, 2014).

This model has been modified by Banker, Charnes, and Cooper (1984) to incorporate “variable returns to scale” by adding a convexity constraint, $\sum_{j=1}^n \lambda_j = 1$ (or $e\lambda = 1$) to the envelopment model of CCR. This new model is also known as BCC model. The BCC model gives the technical efficiency associated with variable returns to scale. The first two decades of work done in the field of DEA was more methodological than application-based (Liu, Lu, Lu, & Lin, 2013). In the last 20 years, there have been numerous studies focusing on the public-sector application of DEA such as healthcare services (Nedelea & Fannin, 2013), national R&D programs (Lee, Park, & Choi, 2009), railways (Emrouznejad, Parker, & Tavares, 2008), nation stability (Tsuneyoshi, Hashimoto, & Haneda, 2012), and police forces (Aristovnik, Seljak, & Mencinger, 2014; Gorman & Ruggiero, 2008).

DEA has also been used to assess the performance of courts. While, there have been efficiency studies across multiple countries (Deyneli, 2012), most studies have been country-specific (Castro & Guccio, 2014; Deyneli, 2012; Falavigna, Ippoliti, & Ramello, 2018; Pedraja-Chaparro & Salinas-Jimenez, 1996; Santos & Amado, 2014; Schneider, 2005; Yeung & Azevedo, 2011). The selection of inputs and outputs is an important concern while conducting DEA analysis. Majority of the work studying judicial efficiency has considered the number of judges (Schneider, 2005), or both, judges and staff as input variables (Castro & Guccio, 2014; Deyneli, 2012; Falavigna et al., 2018; Pedraja-Chaparro & Salinas-Jimenez, 1996; Santos & Amado, 2014; Yeung & Azevedo, 2011). Some studies use caseload as an input (Schneider, 2005), while some consider the number of pending cases (Castro & Guccio, 2014), or both incoming and pending cases as separate variables (Falavigna et al., 2018). The number of cases disposed are considered as the output across most of the literature. The number of outputs differ on account of the number of categories of cases considered. Some consider a single category of cases (Falavigna et al., 2018; Schneider, 2005), while some bifurcate them into multiple categories based on case type (Deyneli, 2012; Santos & Amado, 2014), level of adjudication (Yeung & Azevedo, 2011), or means of resolution (Castro & Guccio, 2014; Pedraja-Chaparro & Salinas-Jimenez, 1996). Some studies incorporate resolution of disputes and judicial lawmaking by considering the number of cases disposed and number of decisions published, respectively, as outputs (Schneider, 2005).

Much of the existing work has taken an output-oriented approach (Falavigna et al., 2018; Santos & Amado, 2014; Schneider, 2005; Yeung & Azevedo, 2011), while a few have considered the input-oriented approach too (Deyneli, 2012; Pedraja-Chaparro & Salinas-Jimenez, 1996). Malmquist Productivity Indices (MPI) (Falavigna et al., 2018) and window analysis (Schneider, 2005) are further extensions of DEA that are used for multi period analysis.

Most of the studies which analyze judicial systems are restricted by national boundaries. This is because, for every judicature, there might be variations in the structure, operating procedures, underlying laws and the environment within which it operates. This prevents generalization of results and raises the need for an independent evaluation of a judicial system. To the best of our knowledge, this is the first ever efficiency analysis of Indian courts. Further, limited work has been done to analyse the impact of incoming cases and pending cases while evaluating the efficiency of courts worldwide. Accordingly, in this paper, we develop DEA models to study the efficiency performance of high courts across India. Further details of the models and corresponding analysis is presented in Section 4.

4. Models and analysis

This section describes the data in Subsection 4.1, the models developed and their results in Subsection 4.2, and a further efficiency analysis of the results in Subsection 4.3.

4.1. Data

The present study aims to analyse the relative efficiency of high courts, the second tier of Indian judiciary. Our study incorporates all of the 24 high courts present in India during the period corresponding to the analysis (2015–18). Note that the current number of high courts in India is 25 (as on 9 April, 2020) because the Hyderabad-based joint High Court of Judicature of the states of Andhra Pradesh and Telengana was bifurcated into two independent high courts on 1 January, 2019. The description of the variables used in the study are presented in Table 2. It is recommended that an isotonic relationship be established between the inputs and outputs before applying DEA (Golany & Yaakov, 1989). Accordingly, a correlation analysis is conducted and the correlations are reported in Table 2. All the correlations are significant at a 95% confidence level. For sake of brevity, only the correlation coefficients for the year 2017–18 are presented in Table 2, although the correlations are computed for the other years (2015–16, 2016–17) as well and found to be statistically significant at a 95% confidence level.

Even though the data is organised into 21 categories of cases (13 civil and 8 criminal), the courts condense them into two categories, namely, civil and criminal. This helps in our DEA analysis as well for the following reasons: firstly, as the number of variables in a DEA model increases, the probability of the model having low discriminatory power rises. Specifically, Friedman and Sinuany-Stern (1998) states that the combined number of inputs and outputs should not exceed one-third the number of DMUs.

There are quantitative and qualitative aspects of evaluating the efficiency of a judicial system. The quantitative aspect often uses measures such as number of cases disposed or clearance rate. The other aspect is qualitative, used to analyse whether the procedures and judgements in court are fair and just. This study does not explore the fairness of judgements. However, judicial efficiency and quality are not mutually exclusive, and a court that provides ‘quality’ decisions, but takes a long time to do so cannot be deemed efficient (Ostrom & Hanson, 1999). Yeung and Azevedo

Table 2
Description of variables.

S.no.	Variable name	Explanation	Input/output	Correlation with disposed – civil	Correlation with disposed – criminal
1.	Judges	Working strength of judges	Input	0.90	0.79
2.	Staff	Working strength of staff	Input	0.86	0.75
3.	Instituted – civil	Number of civil cases instituted during the year	Input	0.95	0.68
4.	Instituted – criminal	Number of criminal cases instituted during the year	Input	0.84	1.00
5.	Caseload – civil	Sum of civil cases pending at the beginning of the year and those instituted during the year	Input	0.97	0.76
6.	Caseload – criminal	Sum of criminal cases pending at the beginning of the year and those instituted during the year	Input	0.85	0.96
7.	Disposed – civil	Number of civil cases disposed during the year	Output	NA	NA
8.	Disposed – criminal	Number of criminal cases disposed during the year	Output	NA	NA

Table 3
Descriptive statistics of the variables (2017–18).

Variable	Total	Mean	Std.	Max	Min
Judges	671	27.96	22.58	93	2
Staff	31534	1313.92	943.50	3762	149
Cases instituted – civil	1024053	42668.88	36574.58	127767	145
Cases instituted – criminal	809065	33711.04	39485.90	175089	71
Caseload – civil	4034489	168103.71	162274.99	675631	272
Caseload – criminal	1996681	83195.04	111753.22	546621	116
Disposed – civil	890266	37094.42	32874.88	140141	98
Disposed – criminal	721136	30047.33	35861.02	156773	47

(2011) supports this view, stating that judicial quality is taken into account through an efficiency analysis, even though it is limited to quicker processing of cases.

The data corresponds to a three-year period from 2015 to 2018. Data is collated from the annual reports ([The Supreme Court of India, 2016, 2017, 2018](#)) published each year by the Supreme Court of India. Each year is defined as starting from 1 July of the year to 30 June of the following year. For example, year 2017–18 refers to the period from 1 July, 2017 to 30 June, 2018. The descriptive statistics of the variables for 2017–18 are presented in [Table 3](#).

4.2. DEA models and results

This section presents the description and results of the three DEA models. Model 1, which considers judicial resources as inputs is presented in Subsection 4.2.1. The impact of caseload per judge on efficiency scores as per Model 1 is discussed in Subsection 4.2.2. The details of Models

2 and 3, which incorporate the number of cases instituted and caseload, respectively, are presented in Subsection 4.2.3. The intent of this study is not to compare the three models, or choose the best among them. Instead, the models are developed to reveal better insights and help decision makers implement targeted reforms by identifying specific areas of inefficiency.

4.2.1. Model 1

4.2.1.1. Model parameters. The first step towards using DEA is to determine appropriate inputs and outputs which are relevant from a policy point of view, and have a statistical relationship with each other. In the first DEA model, referred to as Model 1, we consider two inputs, viz., judges and staff, and two outputs, viz., disposed – civil and disposed – criminal. The use of these variables has been justified in the existing literature (Deyneli, 2012; Falavigna et al., 2018; Pedraja-Chaparro & Salinas-Jimenez, 1996; Santos & Amado, 2014; Schneider, 2005; Yeung & Azevedo, 2011). The efficiency of Indian high courts has not been studied to the best of our knowledge and Model 1 serves as a good starting point to measure their performance.

Output-oriented, variable returns to scale (BCC) models are used to determine technical efficiencies for each year. Models are chosen to be output-oriented since the goal is to maximize the number of cases disposed given the available level of resources and huge backlogs. Further, variable returns to scale are used since the courts show significant variations in size, as seen in Table 3. The DEAP software (Coelli, 1996) is used to run the models and EMS (Scheel, 2000) to verify the results.

4.2.1.2. Results and discussion. We consider a court to be efficient if it has a technical efficiency of 100.0%, and is used as a benchmark for more than one DMU. This is in line with Pedraja-Chaparro and Salinas-Jimenez (1996), according to which, a court is not “genuinely efficient” if it is not considered as a role model for many peers and Cooper, Seiford, and Tone (2007), according to which an efficient DMU that does not appear as an efficient peer a few times needs to be assessed further.

The results of Model 1 are presented in Table 6. Less than 30% of the courts are found to be efficient each year, with the average technical efficiency scores ranging from 70 to 79%. Model 1 considers efficiency in terms of the number of cases disposed per judge and staff. Delhi has the maximum potential for improvement with an average technical efficiency of 38% over the years, probably on account of low disposals per judge (around 1200 per judge). Since Delhi has consistently low efficiency, a further analysis of the appeal rate of the court’s decisions is in order. For example, a slow processing of cases, but low appeal rate could imply that the court is spending higher than average time per case and generating fairer and more acceptable judgements. The worst performing courts for the most recent period (2017–18), their efficient peers, and the weights assigned to them are presented in Fig. 3. The vertical axis in the figure includes the four courts with the lowest efficiency and the horizontal bars represent the corresponding efficient peers along with their weights. For example, Manipur is an inefficient court which has Meghalaya, Orissa, and Uttarakhand as its role models with weights of 0.91, 0.06, and 0.04, respectively.

Model 1 includes only the variables pertaining to staff and judge strength. Other variables such as the number of cases being dealt with are also important inputs beyond the exclusive control of the courts. Further, as seen in Table 3, high courts in India have high variation in the variables. Allahabad has the largest sanctioned strength of judges and staff, at 160 and 4855, respectively, while Sikkim has only 3 and 176, respectively. Similarly, judicial vacancies vary from 16% (Madras) to 53% (Jammu and Kashmir), as indicated in Fig. 1. As such, scientific analysis of the court efficiency often gets overshadowed by generalistic ideas. For example, the only variable

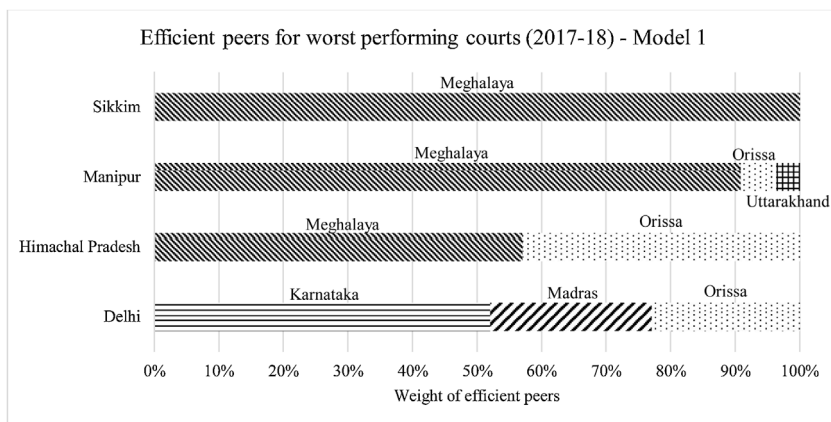


Fig. 3. Efficient peers for worst performing courts (2017–18) – Model 1.

most stakeholders point to for explaining court inefficiency is low judge strength. Our DEA-based analysis goes further. While judge strength clearly is an important variable influencing judicial delay, there is more to it. Consider the case of Allahabad. In 2017–18, the court experienced judicial vacancies in the range of 31–49%. It had the largest caseload (1,222,252 cases), accounting for one-fifth of the caseload across all high courts. Each judge had an average caseload of over 13,000 cases, which was 1.5 times the national average. Despite these numbers, Allahabad was found efficient in all three years. The caseload per judge varied from 15,443 (Orissa) to 194 (Sikkim). Despite a stark difference, Orissa is efficient across all years, while Sikkim is efficient in none. These discrepancies arise because Model 1 considers only the resources available and does not take into account the stress on the judicial system in terms of the number of cases entering the system. It therefore results in the need for a more detailed analysis of these courts' performance, particularly considering the important input of number of cases.

4.2.2. Impact of caseload per judge on efficiency

Before incorporating input variables corresponding to number of cases, we assess the relationship between the new variables and technical efficiency scores from Model 1. A simple analysis of our data reveals that a positive and significant relationship exists (at 95% confidence level) between caseload per judge and technical efficiency, for each year except 2016–17. It can be said that as the work per judges rises, the judges probably feel the burden and take measures to improve their performance and hence, increase outputs. This is in line with the conclusions from [Beenstock and Haitovsky \(2004\)](#), according to which increasing number of judges is not effective in improving performance.

To further assess the impact of caseload per judge, we run a Mann–Whitney *U*-test on the results. A Mann–Whitney *U*-test is a non-parametric test to compare two groups of samples wherein the ranks need to be assessed and determine if the samples come from the same population ([Mann & Whitney, 1947](#)). It has been used in DEA studies ([Aida, Cooper, Pastor, & Sueyoshi, 1998](#); [Lee et al., 2009](#)) to compare the performance of DMUs. It has also been used in the case of judiciary to determine the impact of courts' size and composition of work force on efficiency scores ([Santos & Amado, 2014](#)).

Our study incorporates 24 courts for each year. We divide the courts into two groups of 12 each, based on the median of the caseload per judge. The null hypothesis (H_0) states that there is no

Table 4
Results of Mann–Whitney *U*-test.

Year	2015-16		2016-17		2017-18	
	Group 1	Group 2	Group 1	Group 2	Group 1	Group 2
Average technical efficiency (%)	90.28	66.73	84.95	71.08	84.03	56.90
Mean rank	16.00	9.00	14.21	10.79	16.29	8.71
<i>U</i> -statistic	30.00		51.50		26.50	
p-Value	0.014		0.242		0.007	
Conclusion	Reject H_0		Unable to reject H_0		Reject H_0	

difference in the efficiency distributions of both groups. The test is conducted on SPSS software. The results of the test are presented in Table 4.

The results indicate there is a significant relationship between technical efficiency and caseload per judge for a significance level of 0.05. For the years 2015–16 and 2017–18, the courts with a higher caseload per judge perform better than the courts with caseload per judge lower than the median.

The relationship between efficiency scores and caseload per judge emphasizes on the need to include caseload while measuring court efficiency, as opposed to the Model 1 approach of only considering the personnel (judges and staff) as inputs. Recall from Section 1 that the caseload incorporates cases pending at the beginning of the year as well as fresh cases instituted during the year. Further, there is a need to evaluate the efficiency of courts by considering only fresh cases instituted (excluding pendency) as an input. This is because using caseload or pending cases as an input can sometimes represent the performance of the court in the previous years, and not the period being analysed (Santos & Amado, 2014). It could present a picture wherein a court was found to be inefficient because of its huge pendency, and not due to fewer disposals. Thus, both the approaches (cases instituted alone as an input and caseload as an input) together can provide a holistic perspective required to make an informed decision regarding policy changes. We develop Model 2 and Model 3, accordingly.

4.2.3. Models 2 and 3

4.2.3.1. Model parameters. Model 2 incorporates the inflow of cases, i.e., the number of civil and criminal cases instituted during the year being evaluated. These refer to the variables: instituted – civil and instituted – criminal. The other variables are the same as Model 1. Model 3 replaces the inputs related to cases instituted in Model 2 with caseload – civil and caseload – criminal. A comparison between Models 2 and 3 helps in differentiating between courts that are efficiently managing the cases being instituted during the period of analysis and those that are efficiently managing their total workload. Table 5 summarises the variables considered in each model.

Similar to Model 1, output-oriented, variable returns to scale models are run. The DEAP software (Coelli, 1996) is used to run the model and EMS (Scheel, 2000) to verify the results.

4.2.3.2. Results and discussion. Similar to Model 1, only the courts that have a technical efficiency of 100.0% and are benchmarks for more than one court are considered to be efficient in Models 2 and 3. The average technical efficiency scores for Model 2 range from 95 to 98% (Table 6). Karnataka and Orissa are the top performers, and technically efficient for each year under consideration. With nearly one third of the courts turning out to be efficient, and high efficiency scores for inefficient courts, the results indicate that the courts are performing well with

Table 5

Variables considered in each model.

	Model 1	Model 2	Model 3
Inputs	Judges Staff	Judges Staff Instituted – civil Instituted – criminal	Judges Staff Caseload – civil Caseload – criminal
Outputs	Disposed – civil Disposed – criminal	Disposed – civil Disposed – criminal	Disposed – civil Disposed – criminal

Table 6

Summary of results for each model.

	2015–16	2016–17	2017–18
Model 1			
Number of efficient courts	7	6	7
Percentage of efficient courts (%)	29.17	25.00	29.17
Average technical efficiency scores (%)	78.51	78.01	70.47
Minimum efficiency score (%)	28.78	35.51	35.16
Model 2			
Number of efficient courts	9	9	7
Percentage of efficient courts (%)	37.50	37.50	29.17
Average technical efficiency scores (%)	97.51	97.66	95.47
Minimum efficiency score (%)	88.00	79.94	80.24
Model 3			
Number of efficient courts	9	8	9
Percentage of efficient courts (%)	37.50	33.33	37.50
Average technical efficiency scores (%)	92.50	91.56	92.48
Minimum efficiency score (%)	71.71	66.99	65.50

respect to incoming cases during each year. It also potentially indicates that the courts are well equipped to deal with such a workload each year, despite such high judicial vacancies (average judicial vacancies in high courts was around 40%). Of course, for each state, this notion needs to be reconciled with the actual clearance rate as well before being used as a valid inference.

Despite contrasting views regarding inclusion of cases instituted and pending to analyse court performance, we believe that to tackle a backlog as huge as the one in India, courts need to take strict measures and adopt practices aimed at reducing it. Accordingly, Model 3 which considers both incoming cases during the year and pending cases at the beginning of the year as inputs, is developed. The results of Model 3 seem to indicate good overall performance by the courts, with at least eight courts being deemed efficient every year (Table 6). The average technical efficiencies are around 92% for each year, and the lowest efficiency scores lie between 65–72%. The good efficiency scores can sometimes also be a manifestation not as much of functional efficiency, but of a lack of scope to increase output due to constrained resources (personnel). However, the efficiency scores from Model 3 show high negative correlation with congestion rates implying, thereby, that the congestion effects are not skewing the efficiency scores for most of the courts, and at the same time, reconciling the idea that the courts that are highly congested are not performing well with respect to their pendency levels. A more elaborate picture and corresponding inferences are presented in Table 9.

Table 7
Results of Friedman test.

	Model 1	Model 2	Model 3
Mean ranks			
2015–16	2.15	2.04	1.94
2016–17	2.10	2.17	2.04
2017–18	1.75	1.79	2.02
p-Value	0.243	0.180	0.878

Table 8
Top performing courts under each model.

High court	Average technical efficiency (%) (2015–18)	Number of years considered efficient	Number of times considered benchmark (2015–2018)
Model 1			
Allahabad	100.00	3	17
Gujarat	84.27	2	15
Karnataka	100.00	3	30
Madras	96.07	2	12
Meghalaya	100.00	3	19
Orissa	100.00	3	33
Model 2			
Allahabad	100.00	2	8
Calcutta	100.00	2	9
Gauhati	97.20	2	6
Karnataka	100.00	3	12
Kerala	100.00	2	5
Madras	100.00	2	6
Orissa	100.00	3	13
Patna	100.00	2	4
Tripura	99.50	2	7
Model 3			
Allahabad	100.00	3	7
Delhi	100.00	2	5
Gujarat	94.27	2	19
Karnataka	100.00	3	26
Madras	100.00	3	15
Meghalaya	100.00	2	7
Orissa	100.00	3	12
Patna	100.00	2	11
Tripura	100.00	2	4

A k-sample dependent sample test is run for each model to test whether there has been any significant change in the efficiencies over the years. The test is conducted on SPSS software and level of significance is 0.05. Friedman test indicates that there has been no significant change in the efficiency scores of the courts over the years (Table 7).

Table 8 presents the top performing courts that are technically efficient for at least two of the three years considered. As per Model 1, six high courts, viz., Allahabad, Gujarat, Karnataka, Madras, Meghalaya, and Orissa are technically efficient in at least two out of three years and are considered as a role model by their peers multiple times. On the other hand, the six courts that

are **performing well** (i.e., technically efficient in at least two out of three years) in both Models 2 and 3 are: Allahabad, Karnataka, Madras, Orissa, Patna, and Tripura. It can be said that these courts are **performing well** not only with respect to incoming cases, but also the overall caseload that they are handling.

It is noteworthy that Patna and Tripura are efficient after incorporating the number of cases dealt with as an input. Further, Gujarat and Meghalaya drop out of the **performing well** list in Model 2 but come back in Model 3. These are interesting observations and reveal the following insights:

- Patna and Tripura appear to be inefficient if one only considers the cases disposed and the extent of the judicial resources (number of judges and staff). However, that is not the complete picture since they seem to have a high number of incoming cases as well as a high caseload, both of which they handle *efficiently*.
- Gujarat and Meghalaya appear to be **performing well** if one only considers the outputs, judicial resources and total caseload. Their performance in handling freshly instituted cases is not as *efficient*.

Three courts, namely, Himachal Pradesh, Hyderabad, and Punjab and Haryana, have been ranked among the three worst performers in both models. These courts are neither efficient at managing their caseload, nor are they able to efficiently handle the incoming cases each year.

4.3. Efficiency analysis

Further, a comparison of the performance of high courts according to Models 2 and 3 for the year 2017–18 is presented in Fig. 4. This comparison for the most recent period can be extremely useful to initiate improvement measures for the Indian judiciary. Judicial officials of inefficient courts can determine the cause of their inefficiencies, whether it is due to high number of cases being instituted or high pendency. They can then focus their efforts accordingly, and recommend policy changes to manage the present inflow or reduce the backlog. There are courts that have shown poor results in both models such as Chhattisgarh, Gujarat, Himachal Pradesh, Hyderabad, Madhya Pradesh, Punjab and Haryana. These are inefficient in handling cases instituted and pending. There are courts that are efficient as per both models, such as Gauhati, Karnataka, Meghalaya, Orissa, Patna. These are the courts that need to be referenced by poor performing courts. It seems that these courts are doing something right, in not just dealing with incoming cases, but also managing their backlog. Their court management procedures and laws specific to the state need to be studied and adopted by other courts. Rajasthan, and Jammu and Kashmir are efficient in Model 2, but inefficient in Model 3. These courts are performing well with respect to the inflow of cases for the given year, but are not able to efficiently manage their pending cases. Such courts need to re-evaluate their performance, identify the procedures followed by their peers in Model 3 to help them reduce their pendency levels. There are certain courts that have an efficiency score of 100%, in one of two models, such as Bombay, Calcutta, Delhi, Jharkhand, Manipur. However, we have not considered them as truly efficient since they act as a benchmark for only themselves or one more court.

A summary of the efficient peers for the worst performers during the year 2017–18 under Model 2, and their weights have been presented in Fig. 5. Fig. 6 presents a comparison between the worst performing courts and the top two benchmarks for each of them. The vertical axis in the figure refers to the four courts with the lowest efficiency and the horizontal bars represent the cases

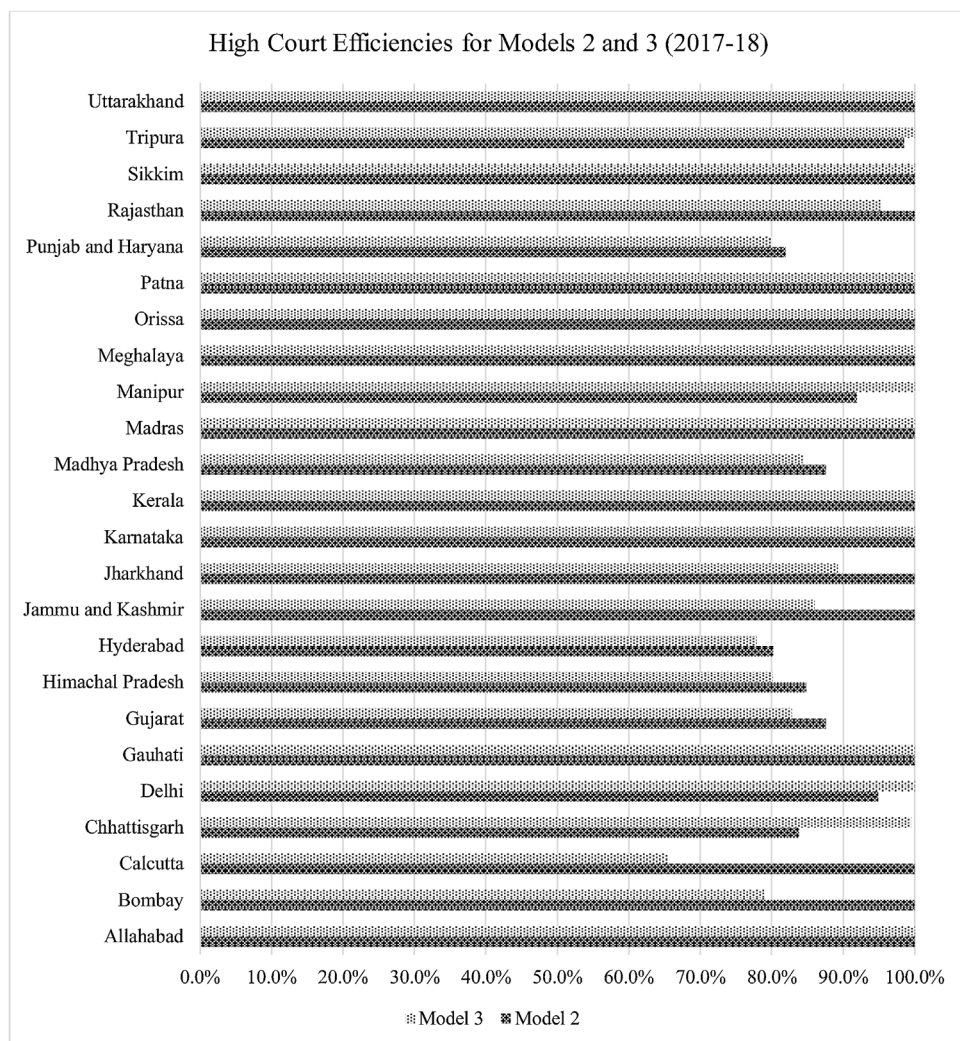


Fig. 4. High court efficiencies for Models 2 and 3 (2017–18).

disposed per judge for the inefficient court, followed by two of its efficient role models having maximum weights. For example, Hyderabad disposed 2216 cases per judge during 2017–18, while its efficient role models, Orissa and Karnataka disposed 5045 and 3039 cases per judge, respectively, for the same time period. The corresponding charts for Model 3 are presented in Figs. 7 and 8. The insights for the inefficient courts that result from a comparative analysis of these figures are presented in Section 5.

Finally, we consider the impact of better utilization of resources on the output of courts. Fig. 9 shows the improvements possible in the variables for each model for the year 2017–18. As per Model 1, number of civil and criminal cases disposed could have increased by 330,432 cases (37%) and 321,198 cases (44%), respectively, had all the courts performed to their full potential. Model 2 shows marginal improvements in the range of 5% for both categories. This implies that

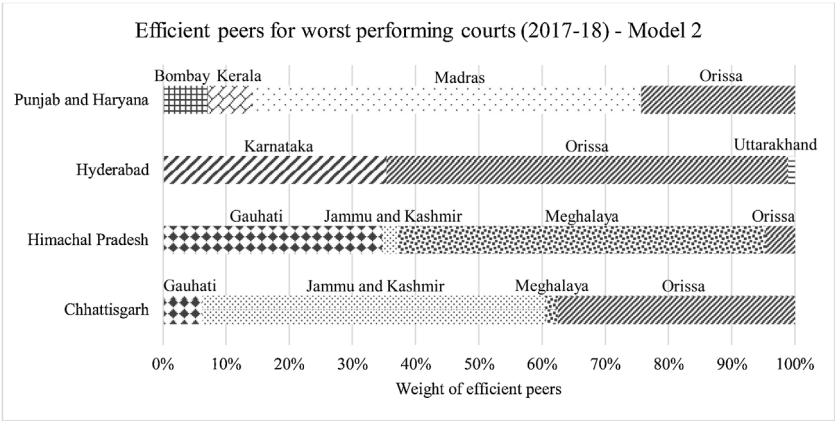


Fig. 5. Efficient peers for worst performing courts (2017–18) – Model 2.

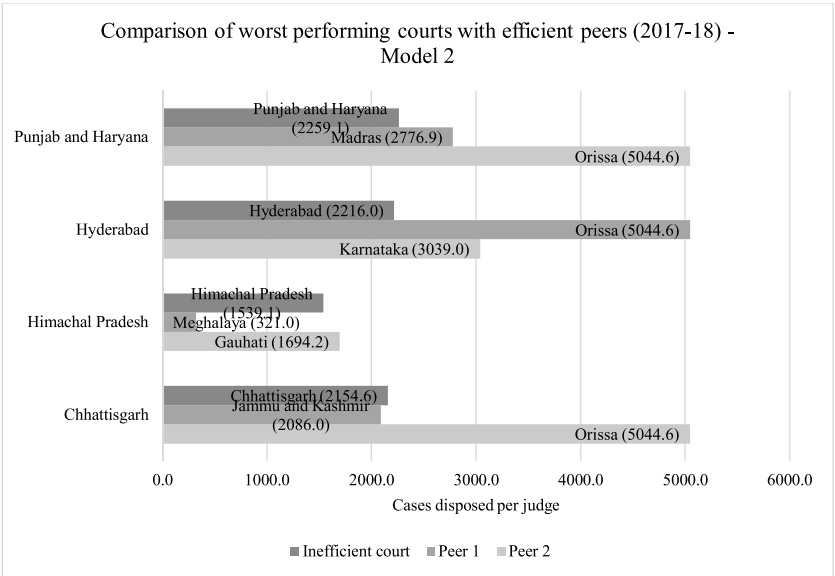


Fig. 6. Comparison of worst performing courts with efficient peers (2017–18) – Model 2.

the courts seem to be performing well in regards to institution of new cases. Model 3 indicates that the outputs could have been increased by 9% for civil and criminal cases. This shows that the courts are capable of performing better and improving their disposal rate, even at their current strength. Judicial vacancies still pose a great threat to a clogged judiciary, but filling these vacancies is not the complete answer. Efforts need to be made to enhance efficiency through effective peer learning and other approaches. Suggested interventions and policy implications are presented in Section 5.

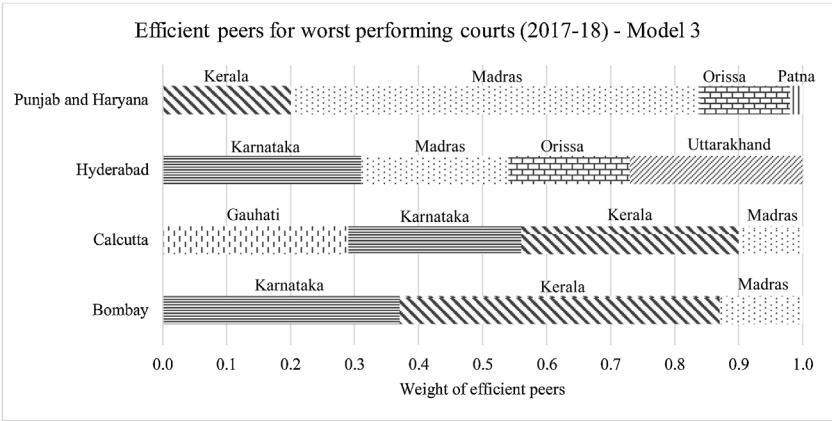


Fig. 7. Efficient peers for worst performing courts (2017–18) – Model 3.

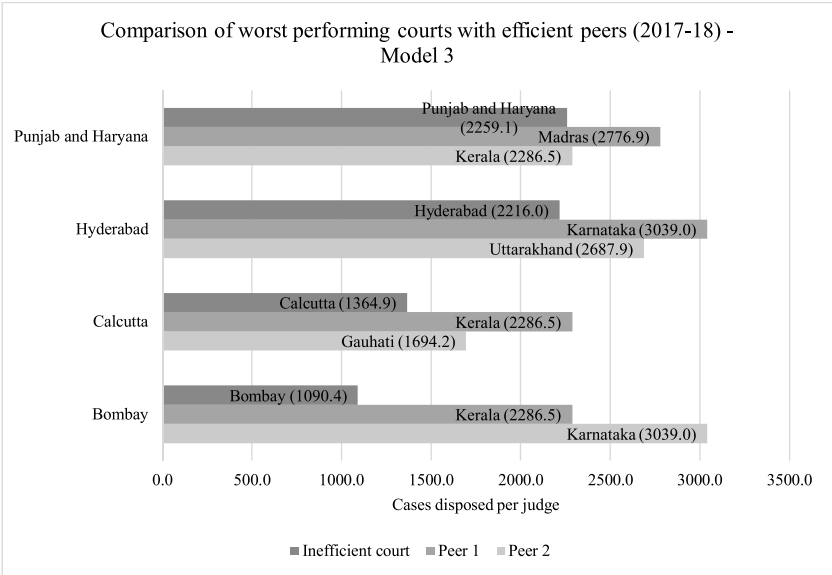


Fig. 8. Comparison of worst performing courts with efficient peers (2017–18) – Model 3.

5. Policy implications

Justice delayed is justice denied. A judicial system has to ensure a timely dispensation of justice, in addition to its judgements being fair and impartial. We have developed three DEA models differing in terms of inputs: the first considers only the judge and staff strength, the second incorporates the cases instituted during the year, and the third considers the caseload. The outputs are the number of cases disposed during the year in all models. While Model 2 presents the courts that are performing well with respect to cases being instituted during the year, Model 3 presents the performance of courts with respect to their overall caseload. Referring to [Tables 6 and 8](#), we can observe that as we move from Model 1 to Models 2 and 3, there is an

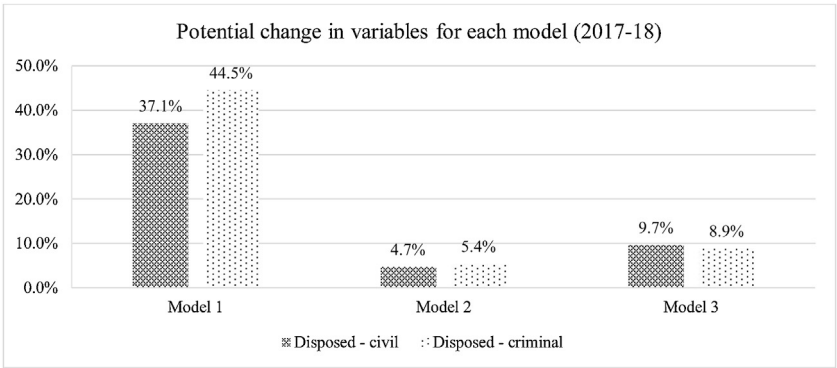


Fig. 9. Potential change in variables (2017–18).

increase in the number of efficient courts and there is no court which is efficient under Model 1, and not under Model 2 or Model 3. It can be inferred that the number of cases instituted and pending is indeed high for Indian high courts, and evaluation of efficiency is incomplete without considering these as inputs and may not yield the full picture.

A comparison of courts across these two models can help determine the specific area where the court is lagging, and can thus enable the court to develop appropriate reforms. Table 9 presents a summary of this comparison and possible policy interventions based on the court’s performance in both models (2017–18). For example, a court efficient under Model 2 but not Model 3, is probably able to manage annual incoming cases but not the total caseload. Such a court should develop policy measures aimed at reducing its pending cases and can focus more on temporary solutions. Similarly, a court efficient as per both models can be used a role model by inefficient courts, its best practices studied and appropriately adapted.

Further, for an inefficient court, its efficient peers and their relative weights may vary according to the model. For example, Hyderabad is inefficient under both Models 2 and 3 (Fig. 4). According to Model 2, its efficient peers are Orissa, Karnataka and Uttarakhand, with relative weights 0.64, 0.35 and 0.01, respectively (Fig. 5). Its efficient peers under Model 3 are Karnataka, Uttarakhand, Madras and Orissa, with relative weights 0.31, 0.27, 0.23 and 0.19, respectively (Fig. 7). It can be observed that Madras is an additional peer in Model 3, and the relative weights of the peers change according to the model. This indicates that an inefficient court needs to consider the model(s) according to which it is performing poorly, determine the specific area where improvements need to be directed, refer to relevant efficient peers, and accordingly develop specific policies.

The study does not present a comparison between the models to determine which model is better. It in fact, presents a comparison between the models to help decision makers understand the area where the court is inefficient. They can then develop specific policies aimed at improving its performance, rather than ambiguous or universal reforms which may not have the desired impact. Accordingly, our models and analysis yield the following policy implications.

5.1. Across the board learnings

Courts such as Allahabad, Karnataka, Madras, and Orissa are *performing well* across all models. Even though these courts have a high volume of cases in terms of instituted and pending, they are also disposing a high number of cases each year. Case management practices of these

Table 9
Suggested policy interventions.

Court efficient in Model 2	Court efficient in Model 3	Example courts (2017–18)	Conclusions about courts	Suggested policy interventions
Yes	No	Jammu and Kashmir, Rajasthan	Courts are able to manage annual incoming cases but not able to manage the total caseload.	• Develop policy measures aimed at reducing its pending cases. • Focus on temporary solutions such as: - Creating ad hoc judge positions - Setting up temporary courts - Better case tracking and management practices to attack backlog.
No	No	Himachal Pradesh, Hyderabad, Madhya Pradesh, Punjab and Haryana	Courts are unable to handle annual incoming cases as well as total caseload.	• Need immediate attention and long-term solutions. • Identify efficient peers, study their policies and practices, and adapt them according to the court's environment. • Focus on measures such as: - Increasing judicial strength - Filling judicial vacancies - Case monitoring systems - Regular monitoring and evaluation techniques.
Yes	Yes	Gauhati, Karnataka, Meghalaya, Orissa	Courts are able to handle annual incoming cases as well as total caseload.	• Be used as role models by inefficient courts, their best practices can be studied and appropriately adapted. - Provide assistance to weak courts in improving their efficiency. • Set up regular monitoring and evaluation techniques to sustain their good performance.
No	Yes	Allahabad, Uttarakhand	Courts are performing well when caseload is considered but not when incoming cases are considered.	• Unable to clear cases at the rate at which they are being instituted, likely to have high pendency. • High DEA score may be due to congestion (no scope to increase output due to limited resources). Increase resources, learn from efficient peers of Model 2.

courts can be studied and adopted since they are managing their yearly inflow of cases, while simultaneously focusing on reducing pendency.

5.2. Specific efficiency learnings

Patna is an example of a court burdened by a high volume of cases, *performing well* according to Models 2 and 3, and not Model 1 (Table 8). This highlights the necessity of including number of cases instituted and pending while measuring efficiency. Similarly, courts such as Calcutta and Kerala are *performing well* under Model 2, but not Models 1 and 3. These courts are efficient with respect to the annual inflow of cases, but not able to manage their pending cases. These courts need to develop reforms aimed at reducing their backlog of cases. Once they are able to tackle the backlog, the courts can continue to operate smoothly and dispose cases at the rate at which they are being instituted. Gujarat and Meghalaya are *performing well* according to Models 1 and 3, but not Model 2. These courts are likely to have high pendency and congestion. They need to focus on increasing their judicial resources and learn best practices from their efficient peers.

5.3. Ideas for systemic interventions

To deal with the inefficiencies related to caseload, courts need to undertake measures to focus on reducing the backlog. Interestingly, even with the current working strength, courts show high technical efficiency. Thus, filling judicial vacancies is not a complete solution to the problem of pendency. While enhanced judge strength through several measures has been proposed by judicial experts (Datar, 2019; Thankur, 2019), other systematic interventions have rarely been considered seriously. For instance, special court management procedures can be set up to resolve cases. High courts can analyse the pending cases to determine the likelihood of cases being disposed, and divide the cases based on the number of probable hearings needed to resolve the case. The cases with a smaller number of hearings can be grouped together and set for hearing on a specific date, and special courts can be set up for this purpose. For every inefficient court, our models have identified efficient peers, along with a weightage for each peer. The high courts can use these conclusions to get a better understanding of their current performance, and get insights on changes required in their operating procedures, internal policies and structure based on learning from efficient peers. They can then adapt those practices that seem relevant and improve their own performance.

We conclude this paper with a summary of all findings and possible future extensions.

6. Conclusions

This section presents a summary of the conclusions from the models developed in Subsection 6.1 and the way forward in Subsection 6.2.

6.1. Summary of findings

In the current study, three DEA models have been developed to measure the efficiency of 24 Indian high courts across a three-year period. In Model 1, we consider the number of working judges and staff as the inputs, and number of civil and criminal cases resolved during the year as outputs. This is in line with the existing literature and serves as a good starting point to measure the efficiency of Indian high courts. Less than 30% of the courts are found efficient for each year,

and the civil and criminal cases disposed could have increased by 37% and 44%, respectively, had the courts been efficient. However, Model 1 considers only the resources available and does not take into account the stress on the judicial system in terms of the number of cases entering the system.

To incorporate other factors that influence the output, the connection between caseload and the Model 1 efficiency is analysed. A Mann–Whitney *U*-test confirms that courts with a high caseload per judge perform better than courts with a low caseload per judge. This could imply that judges in courts with high caseload per judge end up performing better to reduce pending cases. Hence, Models 2 and 3 are developed to take into consideration the number of incoming and pending cases.

Model 2 determines the efficiency considering only incoming cases and Model 3 measures the efficiency based on the caseload. Results from these models indicate that, broadly, courts are handling the cases being instituted each year well (average efficiency 97%, minimum efficiency 80%), but the aggregate performance comes down when it comes to their total caseload (average efficiency 92%, minimum efficiency 66%). A comparison between the models allows us to determine the whether a court is able to handle its annual inflow of cases and is only burdened by high pendency figures, or is it unable to manage even the cases being instituted each year. This, in turn, can help stakeholders develop and implement policy reforms aimed at improving the court's performance in specific areas where it is underperforming.

Additionally, for the year 2017–18, there are certain courts that are performing well on both counts. These include the high courts of Gauhati, Karnataka, Meghalaya, Orissa, Patna. On the other hand, courts such as Chhattisgarh, Gujarat, Himachal Pradesh, Hyderabad, Madhya Pradesh, Punjab and Haryana are not efficient in either. Further, the model results can be used to a) estimate the potential increase in cases disposed if all courts perform efficiently, and b) indicate the role model peers to each court that performs below 100% efficiency.

6.2. *Way forward*

These models can be extended to study the efficiency of district courts across the country. The courts can also be categorised based on their location in terms of rural and urban, or as civil and criminal, and be evaluated separately to measure differences in efficiencies. The complexity associated with different types of cases can be considered, and cases can be categorised according to the effort and time required for their processing. So far, only quantitative aspects of performance have been considered through number of cases instituted and disposed. Although such data is not available currently, judicial efficiency can also be studied by considering qualitative aspects such as judgements of courts, the quality of these judgements, number of times a given judgement has been cited, and proportion of cases for which appeals have been filed in higher courts (Coviello, Ichino, and Persico (2015) has used percentage of appeals as a measure of judicial quality), once such data is publicly available or can be procured.

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